



INFERENCEAL STATISTICS

Project Report

Project Details:

Project Name	Inferential Statistics - Data Analysis
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1. Problems and Analysis

a. PROBLEM 1

A physiotherapist with a male football team is interested in studying the relationship between foot injuries and the positions at which the players play from the data collected.

	Striker	Forward	Attacking Midfielder	Winger	Total
Players Injured	45	56	24	20	145
Players Not Injured	32	38	11	9	90
Total	77	94	35	29	235

Problem statement:

Above is a contingency table which gives the relationship between foot injuries and the positions at which the players play.

Formulae used:

$$P(A) = m/n$$

Probability of event A = Ratio of number of ways that are favourable to occurrence of A/Total number of outcomes of the experiment

For mutually exclusive events:

$$P(A \cup B) = P(A) + P(B)$$

For not mutually exclusive events:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Probability of AUB = Sum of probability of A and B

For independent events:

$$P(A \cap B) = P(A) * P(B)$$

For non-independent events:

$$P(A \cap B) = P(A) * P(B/A)$$

Where $P(A \cap B)$ is Joint Probability and $P(A)$ is Marginal Probability

Conditional Probability:

$$P(A|B) = (P(B|A) * P(A)) / P(B)$$

Questions:

1.1 What is the probability that a randomly chosen player would suffer an injury?

Total players = 235; Players injured = 145; Probability = Players injured/Total players

Answer: There is 0.670 (61.70%) probability that a randomly chosen player would suffer an injury.

1.2 What is the probability that a player is a forward or a winger?

Forward players = 94; Winger players = 29; Total players = 235

Probability = ((Forward player/Total players) + (Winger player/Total players))

Answer: There is 0.523 (52.340%) probability that a player is a forward or a winger.

1.3 What is the probability that a randomly chosen player plays in a striker position and has a foot injury?

Total striker player = 77; Striker injured = 45; Total players = 235

Probability = (Total striker player/Total players) *(Striker injured/Total striker player)

Answer: There is a 0.1914 (19.14%) probability that a randomly chosen player plays in a striker position and has a foot injury.

1.4 What is the probability that a randomly chosen injured player is a striker?

Total players injured = 145; Striker players injured = 45

Probability = Striker players injured / Total players injured

Answer: There is a 0.310 (31%) probability that a randomly chosen injured player is a striker.

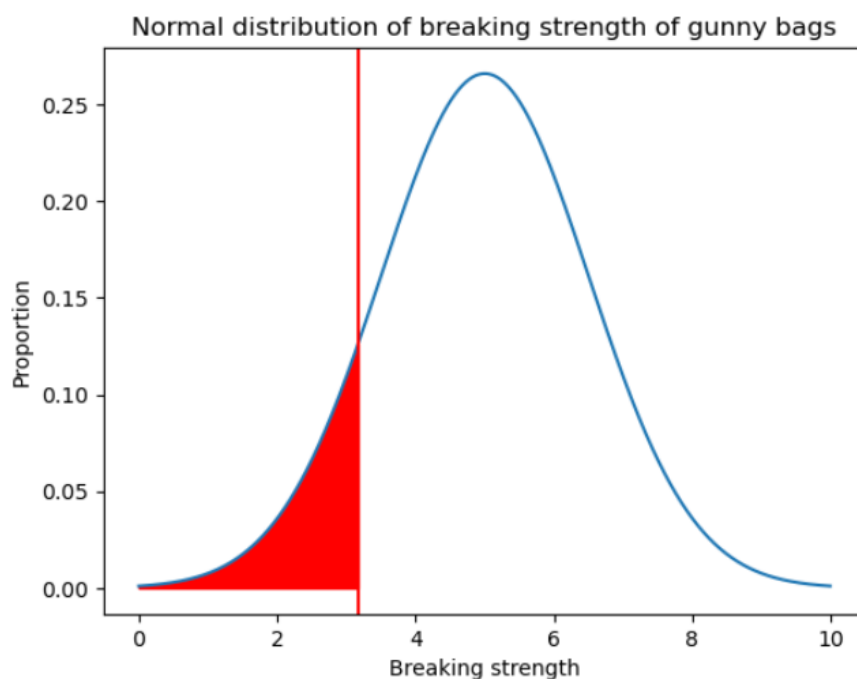
b. PROBLEM 2

The breaking strength of gunny bags used for packaging cement is normally distributed with a mean of 5 kg per sq. centimetre and a standard deviation of 1.5 kg per sq. centimetre. The quality team of the cement company wants to know the following about the packaging material to better understand wastage or pilferage within the supply chain. (Provide an appropriate visual representation of your answers)

Problem statement:

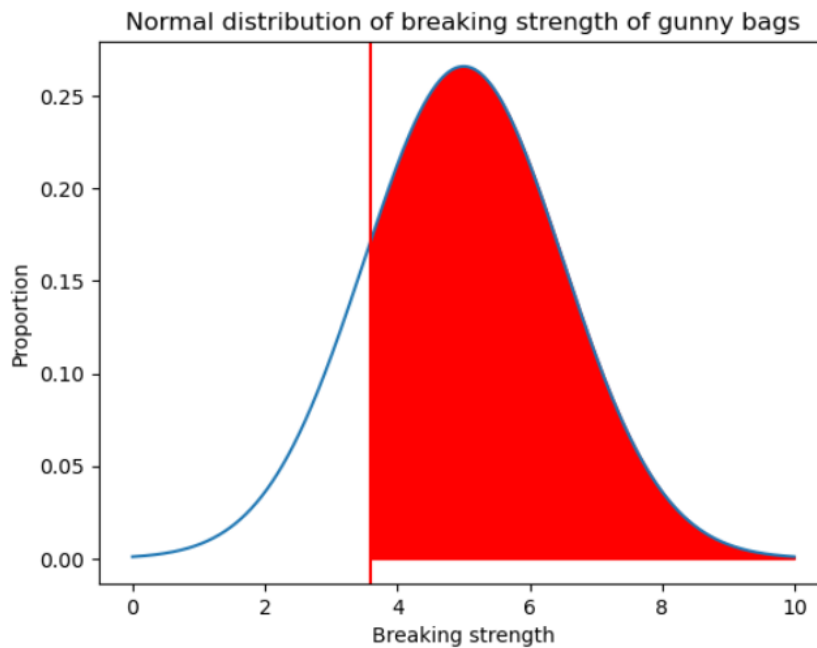
- Population mean $\mu = 5$ kg per sq. centimetre
- Population standard deviation $\sigma = 1.5$ kg per sq. centimetre
- Population distribution = Normal distribution
- Importing numpy, pandas, seaborn, matplotlib libraries for reading data, data computation, data manipulation and data visualization.
- Importing scipy.stats and norm for probability distributions as well as a growing library of statistical functions.

2.1 What proportion of the gunny bags have a breaking strength of less than 3.17 kg per sq cm?



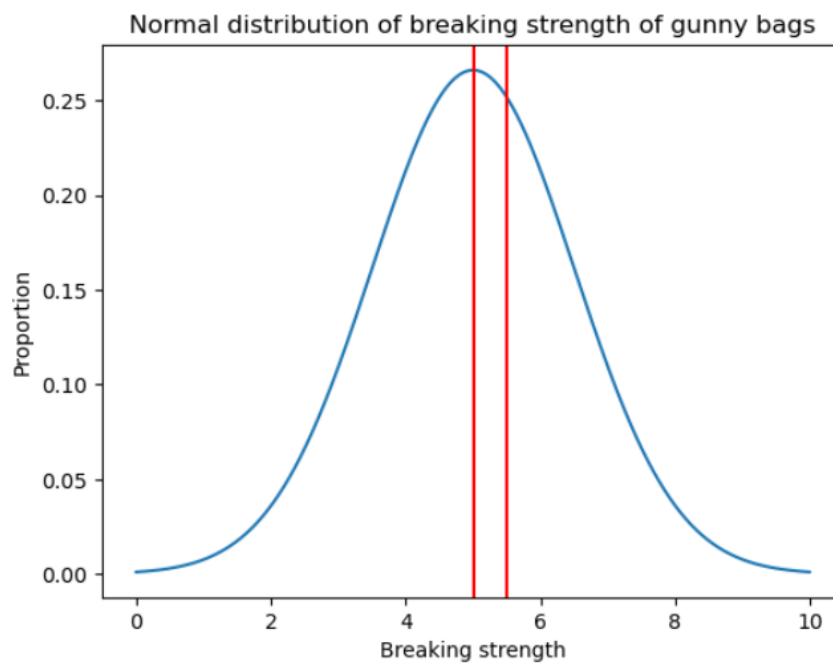
Answer: About 0.1112 (11.12%) of the gunny bags have a breaking strength of less than 3.17 kg per sq cm.

2.2 What proportion of the gunny bags have a breaking strength of at least 3.6 kg per sq cm.?



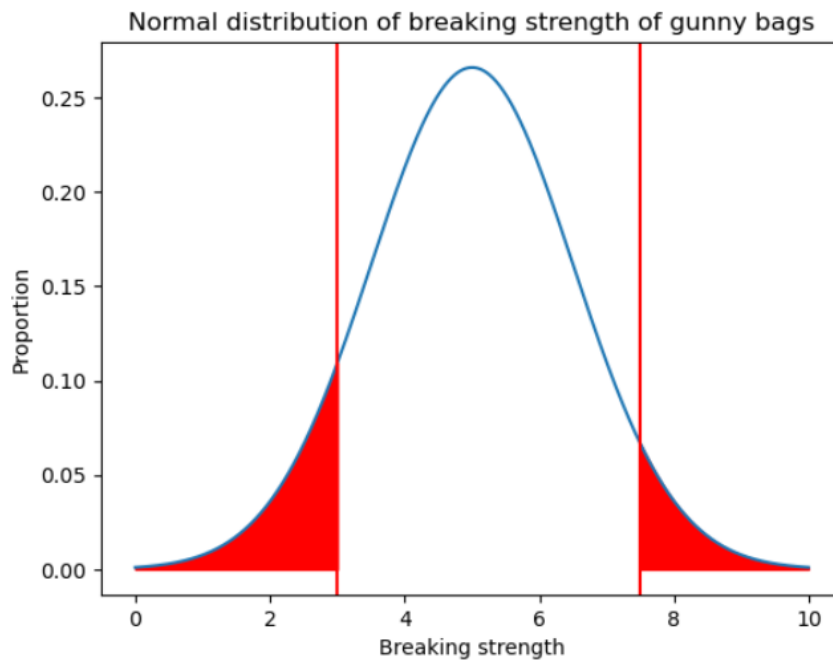
Answer: Almost 0.8247 (82.5%) of the gunny bags have a breaking strength of at least 3.6 kg per sq cm.

2.3 What proportion of the gunny bags have a breaking strength between 5 and 5.5 kg per sq cm.?



Answer: 0.1306 (13%) of the gunny bags have a breaking strength between 5 and 5.5 kg per sq cm.

2.4 What proportion of the gunny bags have a breaking strength NOT between 3 and 7.5 kg per sq cm.?



Answer: 0.1389 (13.9%) of the gunny bags have a breaking strength not between 3 and 7.5 kg per sq cm.

c. PROBLEM 3

Zingaro stone printing is a company that specializes in printing images or patterns on polished or unpolished stones. However, for the optimum level of printing of the image, the stone surface has to have a Brinell's hardness index of at least 150. Recently, Zingaro has received a batch of polished and unpolished stones from its clients. Use the data provided to answer the following (assuming a 5% significance level).

Background:

Brinell's hardness is a unit of measurement that indicates the hardness of a material. Zingaro stone printing seeks to assess the Brinell hardness of both polished and unpolished stones to determine suitable materials for printing images or patterns.

Objective:

To derive conclusions regarding the Brinell's hardness of both polished and unpolished stones and how they vary or align with each other

Data description:

	Unpolished	Treated and Polished
0	164.481713	133.209393
1	154.307045	138.482771
2	129.861048	159.665201
3	159.096184	145.663528
4	135.256748	136.789227

Data structure:

Observation	Variables
75	2

Data types:

- The data set contains two variables indexed from 0-1.
- The data set consists of two numerical columns of float data type.
- There are no null values in the data set.

Missing values:

- No missing values were found in the data set.

Duplicates:

- No duplicates were found in the data set.

Statistical summary:

	count	mean	std	min	25%	50%	75%	max
Unpolished	75.0	134.110527	33.041804	48.406838	115.329753	135.597121	158.215098	200.161313
Treated and Polished	75.0	147.788117	15.587355	107.524167	138.268300	145.721322	157.373318	192.272856

- The mean Brinell's hardness index of Unpolished stones is 134.11.
- The mean Brinell's hardness index of Treated and Polished stones is 147.79.
- The standard deviation of Brinell's hardness index of Unpolished stones is 33.04.
- The standard deviation Brinell's hardness index of Treated and Polished stones is 15.599.
- The means and standard deviations of the two variables are not equal.
- The mean and medians of the variables are almost equal, which suggests that the distribution is a normal distribution.
- There are outliers in the data.

3.1 Zingaro has reason to believe that the unpolished stones may not be suitable for printing. Do you think Zingaro is justified in thinking so?

3.1.1. Null and alternate hypothesis:

Let μ be the mean Brinell's hardness index of Unpolished stones.

We will test the null hypothesis against the alternate hypothesis.

Null hypothesis: The unpolished stones are suitable for printing.

$$H_0 : \mu \geq 150$$

Alternate hypothesis: The unpolished stones are not suitable for printing.

$$H_a : \mu < 150$$

3.1.2. Select appropriate test:

- This is one-tailed test.
- As the population standard deviation is unknown, a one-sample t-test will be appropriate to test the formulated hypotheses.
- The mean and medians of the variables are almost equal, which suggests that the distribution is a normal distribution.
- One sample t-test can be used.

3.1.3. Level of significance: The level of significance (alpha) is 0.05

3.1.4. Calculate P-value:

The P-value is calculated using `ttest_1samp()`.

Test Statistic = -4.164629601426757

p-value = 4.171286997419652e-05

As the p-value is lesser than the level of significance, we reject the null hypothesis. Hence, we have enough significance to conclude that the unpolished stones are not suitable for printing.

3.1.5. Conclusion:

- The mean Brinell's hardness index of unpolished stones is found to be 134.11 from the statistical summary, which contradicts the statement that for optimum level of printing of the image, the stone surface has to have a Brinell's hardness index of at least 150.
- The p-value is lesser than the level of significance, suggesting that we have enough significance to conclude that the unpolished stones are not suitable for printing.
- With the above two evidences, we can conclude that Zingaro is justified in thinking that the unpolished stones are not suitable for printing.

3.2 Is the mean hardness of the polished and unpolished stones the same?**3.2.1. Null and alternate hypothesis:**

Let μ_1, μ_2 be the mean Brinell's hardness index of polished stones and unpolished stones respectively.

We will test the null hypothesis against the alternate hypothesis.

Null hypothesis: The mean Brinell's hardness index of polished stones is equal to the mean Brinell's hardness index of unpolished stones.

$$H_0 : \mu_1 = \mu_2$$

Alternate hypothesis: The mean Brinell's hardness index of polished stones is not equal to the mean Brinell's hardness index of unpolished stones.

$$H_a : \mu_1 \neq \mu_2$$

3.2.2. Select appropriate test:

- This is a two-tailed test.
- As the population standard deviation is unknown, a two-sample t-test will be appropriate to test the formulated hypotheses.
- The mean and medians of the both the polished and unpolished stones are almost equal, which suggests that the distribution is a normal distribution.
- Two sample t-test can be used.

3.2.3. Level of significance: The level of significance (α) is 0.05**3.2.4. Calculate P-value:**

The P-value is calculated using `ttest_ind()`

The p-value is 0.001588379295584306

As the p-value is lesser than the level of significance, we reject the null hypothesis. Hence, we have enough significance to conclude that the mean Brinell's hardness index of polished stones is not equal to the mean Brinell's hardness index of unpolished stones.

3.2.5. Conclusion:

- The mean Brinell's hardness index of unpolished stones and polished stones are found to be 134.11 and 147.79 respectively from the statistical summary, which supports the p-value.
- The p-value is lesser than the level of significance, suggesting that we have enough significance to conclude that the mean Brinell's hardness index of polished stones is not equal to the mean Brinell's hardness index of unpolished stones.
- With the above two evidences, we can conclude that mean hardness of the polished stones is higher than the mean hardness of the unpolished stones.

d. PROBLEM 4

Dental implant data: The hardness of metal implants in dental cavities depends on multiple factors, such as the method of implant, the temperature at which the metal is treated, the alloy used as well as the dentists who may favour one method above another and may work better in his/her favourite method. The response is the variable of interest.

Background:

A dental implant is a prosthesis that interfaces with the bone of the jaw or skull to support a dental prosthesis or act as an orthodontic anchor. Different alloys are being used in the making of the dental implant, which depends on a number of factors such as method of implant, the temperature at which the metal is treated etc.

Objective:

- To derive conclusions regarding the hardness of the implants comprising different alloys.
- To check how different variables contribute to the hardness of the implants.
- To check the interaction between the variables.

Data description:

	Dentist	Method	Alloy	Temp	Response
0	1	1	1	1500	813
1	1	1	1	1600	792
2	1	1	1	1700	792
3	1	1	2	1500	907
4	1	1	2	1600	792

Data structure:

Observation	Variables
90	5

Data types:

- The data set contains five variables indexed from 0-4.
- The data set consists of 2 numerical columns and 3 categorical columns which are shown as integer type.
- There are no null values in the data set.

Missing values:

- No missing values were found in the data set.

Duplicates:

- No duplicates were found in the data set.

Statistical summary:

	count	mean	std	min	25%	50%	75%	max
Dentist	90.0	3.000000	1.422136	1.0	2.0	3.0	4.0	5.0
Method	90.0	2.000000	0.821071	1.0	1.0	2.0	3.0	3.0
Alloy	90.0	1.500000	0.502801	1.0	1.0	1.5	2.0	2.0
Temp	90.0	1600.000000	82.107083	1500.0	1500.0	1600.0	1700.0	1700.0
Response	90.0	741.777778	145.767845	289.0	698.0	767.0	824.0	1115.0

- There are three categorical columns and 2 numerical columns.
- Dentists who utilize dental implants have been classified into five categories.
- Three approaches exist for utilizing dental implants.
- Two distinct alloys are employed in metal implants for dental cavities.
- The average temperature at which the metal is heated is found to be 1600.
- Outliers are absent in the temperature column since the temperature applied to metals is standardized and cannot exceed a specified threshold level.
- The response column has an average of 741 and outliers are seen in the data.
- The mean and median are close, which suggests that the data set follows normal distribution.

Subset the data on the basis of Alloy 1 and Alloy 2

- The dataframe containing data on dental implants, labeled as "df," has been divided into two subsets: "df1" and "df2," in order to address the questions ahead.
- The dataframe df1 comprises Alloy 1 subset.
- The dataframe df2 comprises Alloy 2 subset.

4.1 How does the hardness of implants vary depending on dentists?**4.1.1. ALLOY 1****4.1.1.1 Null and alternate hypothesis:**

Let $\mu_1, \mu_2, \mu_3, \mu_4, \mu_5$ be the means of hardness of implants depending on 5 categories of dentists.

Null hypothesis

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$$

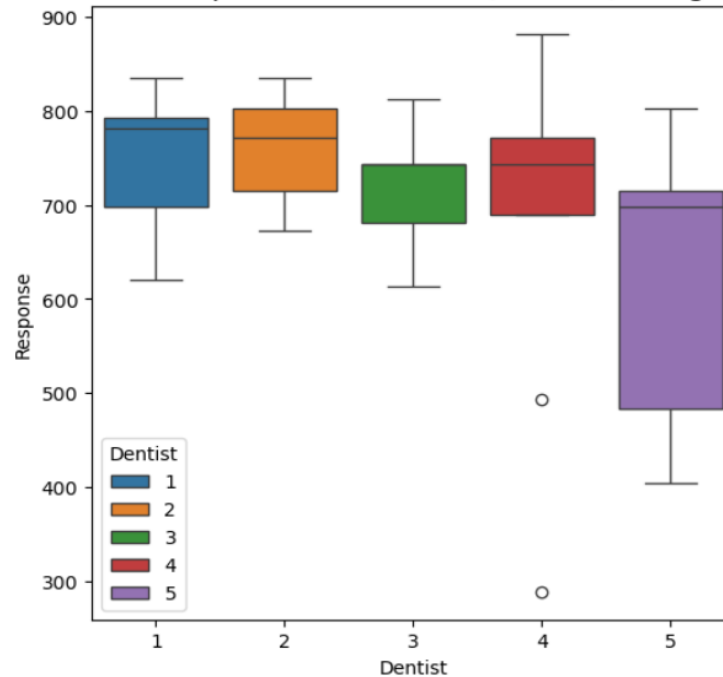
The mean hardness of the implants comprising alloy 1 are same with respect to dentists.

Alternative hypothesis

$$H_a : \text{The mean hardness of the implants comprising alloy 1 vary depending on dentists.}$$

Box plot of means of hardness of implants with respect to 5 categories of dentists using implants comprising alloy 1

Hardness of implant w.r.t. Dentist (5 levels) using ALLOY 1



4.1.1.1 Checking assumptions of hypothesis test

Shapiro-Wilk's test:

Null and alternate hypothesis:

Null hypothesis

H_0 : Hardness of implant follows a normal distribution

Alternative hypothesis

H_a : Hardness of implant does not follow a normal distribution

Calculate P-value:

Assumption: Normality

The P-value is calculated using shapiro()

The p-value is 1.1945070582441986e-05

Since p-value of the test is very small than the level of significance of 0.05, we reject the null hypothesis that the hardness of implant follows a normal distribution.

Insights:

The p-value of suggests that the hardness of implant follows does not follow normal distribution.

Levene's test:

Null and alternate hypothesis:

Null hypothesis

H_0 : All the population variances are equal

Alternative hypothesis

H_a : At least one variance is different from the rest

Calculate P-value:

Assumption:

Homogeneity of Variance

The P-value is calculated using `levene()`

The p-value is 0.2565537418543795

Since the p-value is larger than the level of significance of 0.05, we fail to reject the null hypothesis of homogeneity of variances.

Insights:

The p-value suggests that the hardness of implant follows homogeneity of variances.

Conclusion:

- The populations are not normally distributed. Here, we are assuming it to be.
- Samples are assumed to be independent simple random samples.
- Population variances are equal. The homogeneity of variance assumption is verified using the
- Levene's test.

Multiple Comparison test (Tukey HSD)

Null and alternate hypothesis:

Null hypothesis

H_0 : Hardness of implant is the same for all the dentist groups.

Alternative hypothesis

H_a : Hardness of implant varies among atleast one of the dentist group.

Calculate P-value:

The P-value is calculated using `pairwise_tukeyhsd()`

Multiple Comparison of Means - Tukey HSD, FWER=0.05						
group1	group2	meandiff	p-adj	lower	upper	reject
1	2	11.3333	0.9996	-145.0423	167.709	False
1	3	-32.3333	0.9757	-188.709	124.0423	False
1	4	-68.7778	0.7189	-225.1535	87.5979	False
1	5	-122.2222	0.1889	-278.5979	34.1535	False
2	3	-43.6667	0.9298	-200.0423	112.709	False
2	4	-80.1111	0.5916	-236.4868	76.2646	False
2	5	-133.5556	0.1258	-289.9312	22.8201	False
3	4	-36.4444	0.9626	-192.8201	119.9312	False
3	5	-89.8889	0.4805	-246.2646	66.4868	False
4	5	-53.4444	0.8643	-209.8201	102.9312	False

As the p-values in the p-adj column for comparing the mean hardness of the implants, is higher than the significance level between all groups, we fail to reject the null hypothesis.

Insights:

We can say that the mean hardness of the implants is the same for all the dentist groups. It does not vary depending on the dentists.

4.1.1.2 Select appropriate test:

One-way ANOVA can be used to check if the hardness of implants varies depending on dentists.

4.1.1.3 Level of significance:

The level of significance (alpha) is 0.05

4.1.1.4 Calculate P-value:

The P-value is calculated using `f_oneway()`

The p-value is 0.11656712140267628

C(Dentist)	4.0	106683.688889	26670.922222	1.977112	0.116567
Residual	40.0	539593.555556	13489.838889	NaN	NaN

Since the p-value is greater than the level of significance (5%), we fail to reject the null hypothesis. Hence, we have enough statistical evidence to conclude that the mean hardness of the implants do not vary depending on dentists.

4.1.1.5 Conclusion for ALLOY 1

- As the p-value is higher than the significance level of 0.05, we fail to reject the null hypothesis. Hence, we have enough statistical significance to conclude that mean hardness of the implants comprising alloy 1 does not vary depending on dentists at 5% significance level.
- From the box plot, it is evident that mean hardness of the implants remains same for all the dentist groups, but however the fifth dentist group shows a wider range in the hardness of the implants.

4.1.2. ALLOY 2

4.1.2.1 Null and alternate hypothesis:

Let $\mu_1, \mu_2, \mu_3, \mu_4, \mu_5$ be the means of hardness of implants depending on 5 categories of dentists.

Null hypothesis

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$$

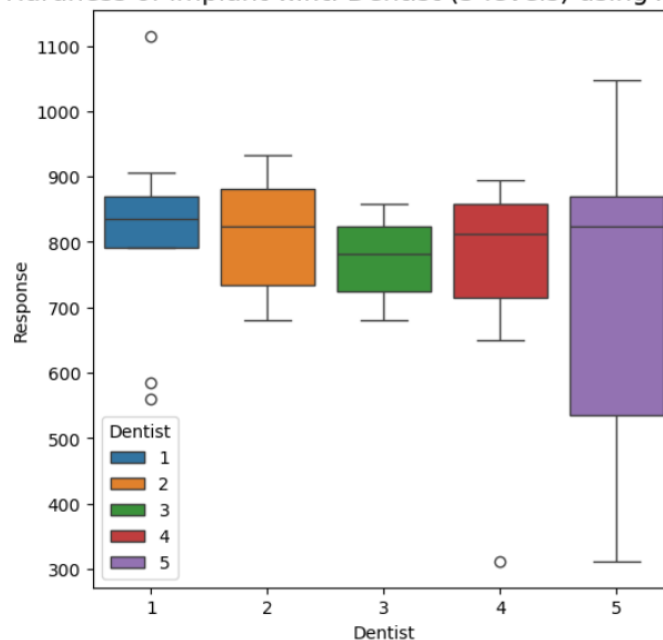
The mean hardness of the implants comprising alloy 2 are same with respect to dentists.

Alternative hypothesis

$$H_a : \text{The mean hardness of the implants comprising alloy 2 vary depending on dentists.}$$

Box plot of means of hardness of implants with respect to 5 categories of dentists using implants comprising alloy 2

Hardness of implant w.r.t. Dentist (5 levels) using ALLOY 2



4.1.2.1 Checking assumptions of hypothesis test

Shapiro-Wilk's test

Null and alternate hypothesis:

Null hypothesis

$$H_0 : \text{Hardness of implant follows a normal distribution}$$

Alternative hypothesis

$$H_a : \text{Hardness of implant does not follow a normal distribution}$$

Calculate P-value:**Assumption:**

Normality

The P-value is calculated using shapiro()

The p-value is 0.00040293222991749644

Since p-value of the test is very small than the level of significance of 0.05, we reject the null hypothesis that the hardness of implant follows a normal distribution.

Insights:

The p-value suggests that the hardness of implant follows does not follow normal distribution.

Levene's test**Null and alternate hypothesis:**

Null hypothesis

H_0 : All the population variances are equal

Alternative hypothesis

H_a : At least one variance is different from the rest

Calculate P-value:**Assumption:**

Homogeneity of Variance

The P-value is calculated using levene()

The p-value is 0.23686777576324952

Since the p-value is larger than the level of significance of 0.05, we fail to reject the null hypothesis of homogeneity of variances.

Insights:

The p-value suggests that the hardness of implant follows homogeneity of variances.

Conclusion:

- The populations are not normally distributed. Here, we are assuming it to be.
- Samples are assumed to be independent simple random samples.
- Population variances are equal. The homogeneity of variance assumption is verified using the
- Levene's test.

Multiple Comparison test (Tukey HSD)

Null and alternate hypothesis:

Null hypothesis

H_0 : Hardness of implant is the same for all the dentist groups.

Alternative hypothesis

H_a : Hardness of implant varies among atleast one of the dentist group.

Calculate P-value:

The P-value is calculated using `pairwise_tukeyhsd()`

Multiple Comparison of Means - Tukey HSD, FWER=0.05						
group1	group2	meandiff	p-adj	lower	upper	reject
1	2	11.3333	0.9996	-145.0423	167.709	False
1	3	-32.3333	0.9757	-188.709	124.0423	False
1	4	-68.7778	0.7189	-225.1535	87.5979	False
1	5	-122.2222	0.1889	-278.5979	34.1535	False
2	3	-43.6667	0.9298	-200.0423	112.709	False
2	4	-80.1111	0.5916	-236.4868	76.2646	False
2	5	-133.5556	0.1258	-289.9312	22.8201	False
3	4	-36.4444	0.9626	-192.8201	119.9312	False
3	5	-89.8889	0.4805	-246.2646	66.4868	False
4	5	-53.4444	0.8643	-209.8201	102.9312	False

As the p-values in the p-adj column for comparing the mean hardness of the implants, is higher than the significance level between all groups, we fail to reject the null hypothesis.

Insights:

We can say that the mean hardness of the implants is the same for all the dentist groups. It does not vary depending on the dentists.

4.1.2.2 Select appropriate test:

One-way ANOVA can be used to check if the hardness of implants varies depending on dentists.

4.1.2.3 Level of significance:

The level of significance (alpha) is 0.05

4.1.2.4 Calculate P-value:

The P-value is calculated using `f_oneway()`

The p-value is 0.7180309510793431

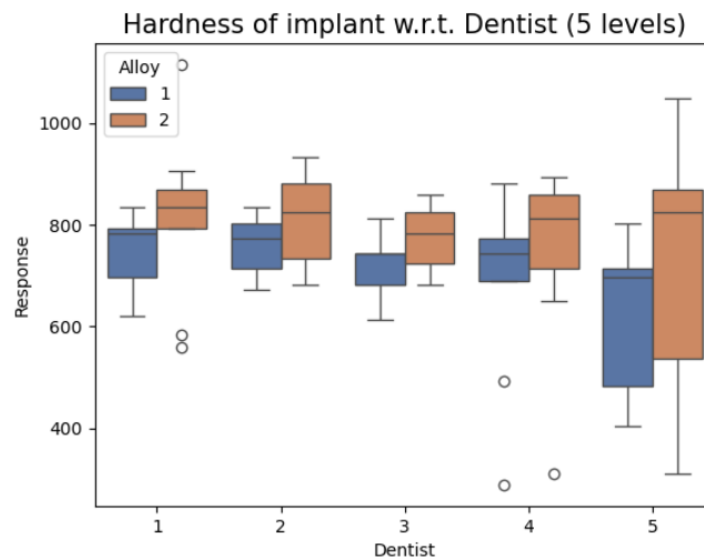
	df	sum_sq	mean_sq	F	PR(>F)
C(Dentist)	4.0	5.679791e+04	14199.477778	0.524835	0.718031
Residual	40.0	1.082205e+06	27055.122222	NaN	NaN

Since the p-value is greater than the level of significance (5%), we fail to reject the null hypothesis. Hence, we have enough statistical evidence to conclude that the mean hardness of the implants does not vary depending on dentists.

4.1.2.5 Conclusion for ALLOY 2

- As the p-value is higher than the significance level of 0.05, we fail to reject the null hypothesis. Hence, we have enough statistical significance to conclude that mean hardness of the implants comprising alloy 2 does not vary depending on dentists at 5% significance level.
- From the box plot, it is evident that mean hardness of the implants remains same for all the dentist groups, but however the fifth dentist group shows a wider range in the hardness of the implants.

Box plot of means of hardness of implants comprising alloy 1 or alloy 2 with respect to 5 categories of dentists



4.1.3 CONCLUSION:

- The mean hardness of the implants does not vary depending on dentists, for both alloy 1 and alloy.
- The hardness of the implant is independent of the dentist.

4.2 How does the hardness of implants vary depending on methods?

4.2.1. ALLOY 1

4.2.1.1 Null and alternate hypothesis:

Let μ_1, μ_2, μ_3 be the means of hardness of implants depending on 3 methods for dental implant.

Null hypothesis

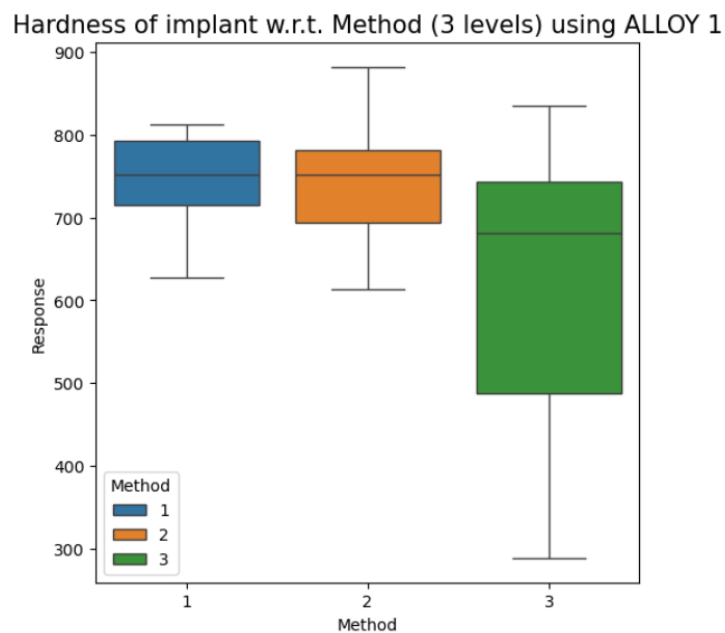
$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The mean hardness of the implants comprising alloy 1 are same with respect to methods for dental implant.

Alternative hypothesis

H_a : The mean hardness of the implants comprising alloy 1 vary depending on methods for dental implant.

Box plot of means of hardness of implants with respect to 3 categories of methods using implants comprising alloy 1



4.2.1.1 Checking assumptions of hypothesis test

Shapiro-Wilk's test

Null and alternate hypothesis:

Null hypothesis

H_0 : Hardness of implant follows a normal distribution

Alternative hypothesis

H_a : Hardness of implant does not follow a normal distribution

Calculate P-value:

Assumption:

Normality

The P-value is calculated using shapiro()

The p-value is 1.1945070582441986e-05

Since p-value of the test is very small than the level of significance of 0.05, we reject the null hypothesis that the hardness of implant follows a normal distribution.

Insights:

The p-value of suggests that the hardness of implant follows does not follow normal distribution.

Levene's test

Null and alternate hypothesis:

Null hypothesis

H_0 : All the population variances are equal

Alternative hypothesis

H_a : At least one variance is different from the rest

Calculate P-value:

Assumption:

Homogeneity of Variance

The P-vale is calculated using levene()

The p-value is 0.0034160381460233975

Since the p-value is lesser than the level of significance of 0.05, we reject the null hypothesis of homogeneity of variances.

Insights:

The p-value suggests that the hardness of implant does not follow the homogeneity of variances.

Conclusion:

- The populations are not normally distributed. Here, we are assuming it to be.
- Samples are assumed to be independent simple random samples.
- Population variances are not equal. Here, we are assuming it to be.

Multiple Comparison test (Tukey HSD)

Null and alternate hypothesis:

Null hypothesis

H_0 : Hardness of implant comprising alloy 1 is the same for all the methods for the implant.

Alternative hypothesis

H_a : Hardness of implant comprising alloy 1 varies among atleast one of the methods for the implant.

Calculate P-value:

The P-value is calculated using `pairwise_tukeyhsd()`

Multiple Comparison of Means - Tukey HSD, FWER=0.05						
group1	group2	meandiff	p-adj	lower	upper	reject
1	2	-6.1333	0.987	-102.714	90.4473	False
1	3	-124.8	0.0085	-221.3807	-28.2193	True
2	3	-118.6667	0.0128	-215.2473	-22.086	True

As the p-values in the p-adj column for comparing the mean hardness of the implants, is higher between method 1 and method 2 than the significance level, we fail to reject the null hypothesis.

However, the mean difference between method 1 and method 2 is slightly higher than the mean difference between method 1 and method 3 and method 2 and method 3. Hence, we reject the null hypothesis.

Insights:

- We can say that the mean hardness of the implants is the same between method 1 and method two.
- The hardness of the implant varies between method 1 and method 3 and method 2 and method 3 depending on the method of implants.

4.2.1.2 Select appropriate test:

One-way ANOVA can be used to check if the hardness of implants varies depending on dentists.

4.2.1.3 Level of significance:

The level of significance (alpha) is 0.05

4.2.1.5 Calculate P-value:

The P-value is calculated using `f_oneway()`

The p-value is 0.004163412167505543

	df	sum_sq	mean_sq	F	PR(>F)
C(Method)	2.0	148472.177778	74236.088889	6.263327	0.004163
Residual	42.0	497805.066667	11852.501587	NaN	NaN

Since the p-value is lesser than the level of significance (5%), we reject the null hypothesis. Hence, we have enough statistical evidence to conclude that the mean hardness of the implants varies depending on methods.

4.2.1.6 Conclusion for ALLOY 1

- We have enough statistical evidence to conclude that the mean hardness of the implants varies depending on methods.
- From the box plot, it is evident that mean hardness of the implants remains same between method 1 and method two.
- However, the mean hardness of the implants is significantly different between method 1 and method 3 and method 2 and method 3 depending on the method of implants.

4.2.2 ALLOY 2

4.2.2.1 Null and alternate hypothesis:

Let μ_1, μ_2, μ_3 be the means of hardness of implants depending on 3 methods for dental implant.

Null hypothesis

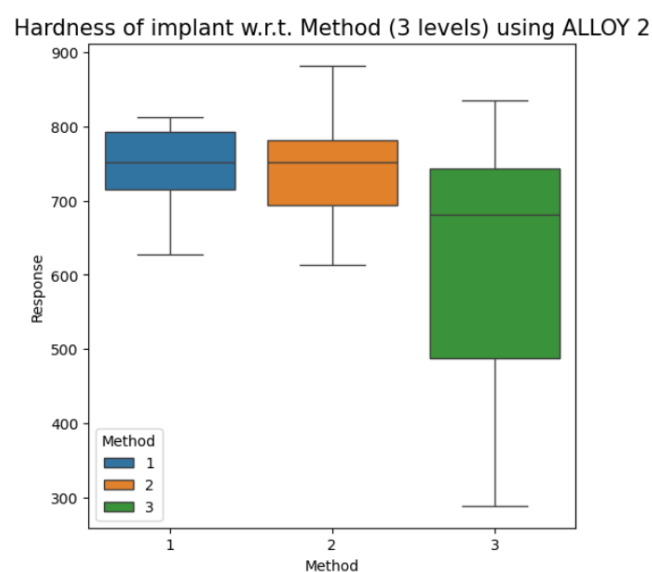
$$H_0 : \mu_1 = \mu_2 = \mu_3$$

The mean hardness of the implants comprising alloy 2 are same with respect to methods for dental implant.

Alternative hypothesis

H_a : The mean hardness of the comprising using alloy 2 vary depending on methods for dental implant.

Box plot of means of hardness of implants with respect to 3 methods using implants comprising alloy 2



4.2.2.2 Checking assumptions of hypothesis test

Shapiro-Wilk's test

Null and alternate hypothesis:

Null hypothesis

H_0 : Hardness of implant follows a normal distribution

Alternative hypothesis

H_a : Hardness of implant does not follow a normal distribution

Calculate P-value:

Assumption:

Normality

The P-value is calculated using shapiro()

The p-value is 1.1945070582441986e-05

Since p-value of the test is very small than the level of significance of 0.05, we reject the null hypothesis that the hardness of implant follows a normal distribution.

Insights:

The p-value suggests that the hardness of implant follows does not follow normal distribution.

Levene's test

Null and alternate hypothesis:

Null hypothesis

H_0 : All the population variances are equal

Alternative hypothesis

H_a : At least one variance is different from the rest

Calculate P-value:

Assumption:

Homogeneity of Variance

The P-value is calculated using levene()

The p-value is 0.04469269939158668

Since the p-value is lesser than the level of significance of 0.05, we reject the null hypothesis of homogeneity of variances.

Insights:

The p-value suggests that the hardness of implant does not follow the homogeneity of variances.

Conclusion:

- The populations are not normally distributed. Here, we are assuming it to be.
- Samples are assumed to be independent simple random samples.
- Population variances are not equal. Here, we are assuming it to be.

Multiple Comparison test (Tukey HSD)**Null and alternate hypothesis:**

Null hypothesis

H_0 : Hardness of implant comprising alloy 2 is the same for all the methods for the implant.

Alternative hypothesis

H_a : Hardness of implant comprising alloy 2 varies among atleast one of the methods for the implant.

Calculate P-value:

The P-value is calculated using `pairwise_tukeyhsd()`

Multiple Comparison of Means - Tukey HSD, FWER=0.05						
group1	group2	meandiff	p-adj	lower	upper	reject
1	2	27.0	0.8212	-82.4546	136.4546	False
1	3	-208.8	0.0001	-318.2546	-99.3454	True
2	3	-235.8	0.0	-345.2546	-126.3454	True

As the p-values in the p-adj column for comparing the mean hardness of the implants, is higher between method 1 and method 2 than the significance level, we fail to reject the null hypothesis.

However, the mean difference between method 1 and method 2 is slightly higher than the mean difference between method 1 and method 3 and method 2 and method 3. We reject the null hypothesis.

Insights:

- We can say that the mean hardness of the implants is the same between method 1 and method two.
- The hardness of the implant varies between method 1 and method 3 and method 2 and method 3 depending on the method of implants.

4.2.2.3 Select appropriate test:

One-way ANOVA can be used to check if the hardness of implants varies depending on dentists.

4.2.2.4 Level of significance:

The level of significance (alpha) is 0.05

4.2.2.5 Calculate P-value:

The P-value is calculated using `f_oneway()`

The p-value is 5.415871051443187e-06

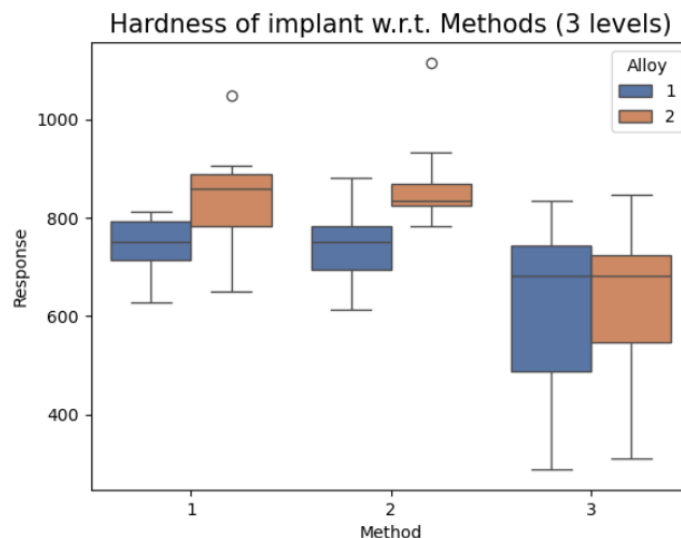
	df	sum_sq	mean_sq	F	PR(>F)
C(Method)	2.0	499640.4	249820.200000	16.4108	0.000005
Residual	42.0	639362.4	15222.914286	NaN	NaN

Since the p-value is lesser than the level of significance (5%), we reject the null hypothesis. Hence, we have enough statistical evidence to conclude that the mean hardness of the implants varies depending on methods.

4.2.1.7 Conclusion for ALLOY 2

- We have enough statistical evidence to conclude that the mean hardness of the implants varies depending on methods.
- From the box plot, it is evident that mean hardness of the implants remains same between method 1 and method two.
- However, the mean hardness of the implants is significantly different between method 1 and method 3 and method 2 and method 3 depending on the method of implants.

Box plot of means of hardness of implants comprising alloy 1 or alloy 2 with respect to 3 methods



4.2.2 CONCLUSION:

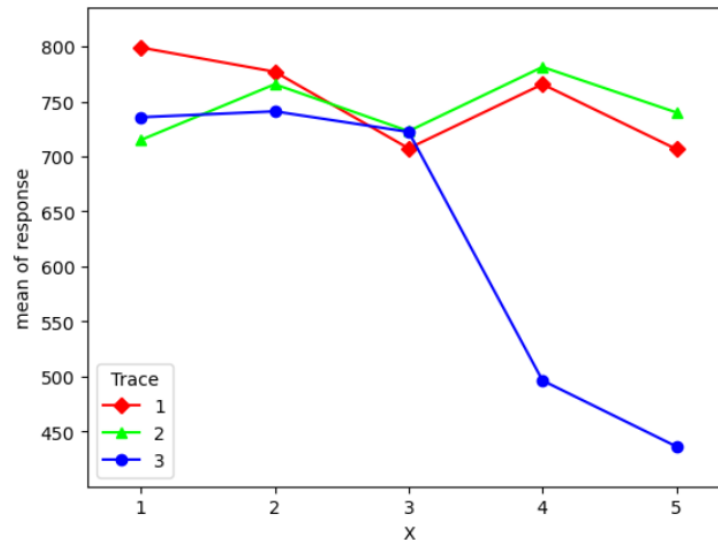
- The mean hardness of the implants varies significantly depending on methods, for both alloy 1 and alloy 2.
- For method 1 and method 2, the mean hardness of the implants remains approximately same for both implants comprising alloy 1 and alloy 2.

- The mean hardness of the implants differs significantly between method 1 and method 3 and method 2 and method 3 depending on the method of implants the mean hardness of the implants.
- The hardness of the implant is dependent on the method for implant.

4.3 What is the interaction effect between the dentist and method on the hardness of dental implants for each type of alloy?

4.3.1 ALLOY 1

4.3.1.1 Interaction plot



4.3.1.2 Inferences from the plot

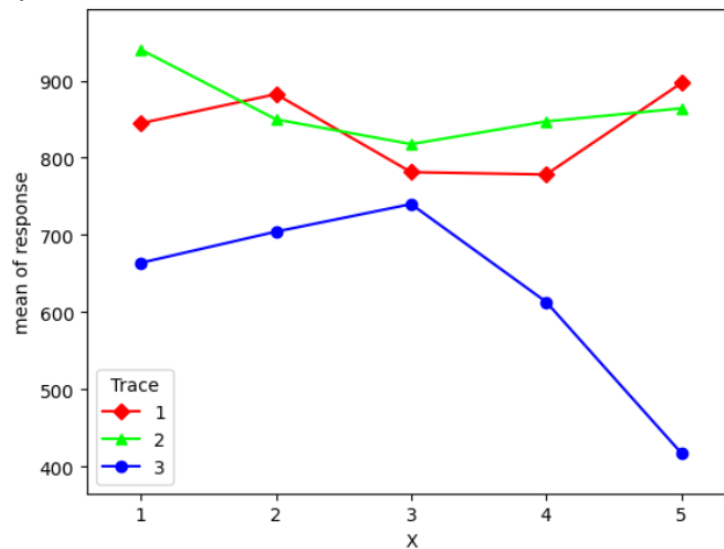
- There are no parallel lines in the plot, implying potential interaction between the dentist and the methods used. This interaction effect indicates that the relationship between dentist and hardness of implant depends on the method of implant used.
- *When dentist 1 employs method 1, the mean response is higher, whereas it drops when method 2 is used.
- When dentist 5 employs method 2, the mean response is higher, whereas it drops when method 3 is used.
- Method 1 shows a higher response with dentist 2, whereas method 3 demonstrates a decrease in response.
- Method 3 shows significantly reduced response when used by dentist 4.
- The lines representing each method intersect at dentist 3, suggesting there may not be significant interaction between the methods associated with dentist 3.
- However, the actual interaction effect between dentist and method can only be confirmed through an ANOVA table.

C(Dentist):C(Method)	8.0	185941.377778	23242.672222	3.398383	0.006793
Residual	30.0	205180.000000	6839.333333	NaN	NaN

*As the p-value is lower than the level of significance of 0.05, we reject the null hypothesis, thereby confirming the presence of interaction effect between dentist and method on hardness of implants comprising alloy 1.

4.3.2 ALLOY 2

4.3.2.1 Interaction plot



4.3.2.2 Inferences from the plot

- There is no evident intersection between all the 3 methods which questions the potential interaction between the dentist and the methods used.
- *When dentist 1 employs method 2, the mean response is higher, whereas it drops when method 3 is used.
- When dentist 5 employs method 1, the mean response is higher, whereas it drops when method 3 is used.
- Method 1 shows a higher response with dentist 2, whereas method 3 demonstrates a decrease in response.
- Method 3 shows significantly reduced response when used by dentist 4.
- The lines representing method 1 and method 2 interact with each other however the method 3 does not.
- At dentist 3, all three methods yield similar responses, indicating there might not be substantial interaction among the methods linked to dentist 3.
- However, the actual interaction effect between dentist and method can only be confirmed through an ANOVA table.

C(Dentist):C(Method)	8.0	197459.822222	24682.477778	1.922787	0.093234
Residual	30.0	385104.666667	12836.822222	NaN	NaN

- *As the p-value is higher than the level of significance of 0.05, we fail to reject the null hypothesis, thereby confirming there is no presence of interaction effect between dentist and method on hardness of implants comprising alloy 2.

4.4 How does the hardness of implants vary depending on dentists and methods together?

4.4.1 ALLOY 1

4.4.1.1 Null and alternate hypothesis:

Null hypothesis

H_0 : The hardness of the implants comprising alloy 1 does not depend on dentists and methods for implants.

Alternative hypothesis

H_a : There is an interaction effect between dentist and method on hardness of the implants.

4.4.1.2 Checking assumptions of hypothesis test

- The problem concerns with the effect of two independent variables on a dependent variable.
- Two-way ANOVA test is the appropriate test.

Following are the assumptions of the Two-way ANOVA test:

- The populations from which the samples are obtained must be normally distributed.

Shapiro-Wilk's test for hardness of implants comprising alloy 1

Null and alternate hypothesis:

Null hypothesis

H_0 : Hardness of implant follows a normal distribution

Alternative hypothesis

H_a : Hardness of implant does not follow a normal distribution

Calculate P-value:

Assumption:

Normality

The P-value is calculated using shapiro()

The p-value is 1.1945070582441986e-05

Since p-value of the test is very small than the level of significance of 0.05, we reject the null hypothesis that the hardness of implant follows a normal distribution.

Insights:

The p-value of suggests that the hardness of implant follows does not follow normal distribution.

- The variances among populations must be equal.

Levene's test for hardness of implants comprising alloy 1 with respect to dentists

Null and alternate hypothesis:

Null hypothesis

H_0 : All the population variances are equal

Alternative hypothesis

H_a : At least one variance is different from the rest

Calculate P-value:

Assumption:

Homogeneity of Variance

The P-value is calculated using levene()

The p-value is 0.2565537418543795

Since the p-value is larger than the level of significance of 0.05, we fail to reject the null hypothesis of homogeneity of variances.

Insights:

The p-value suggests that the hardness of implant follows homogeneity of variances.

Conclusion:

- Population variances are equal. The homogeneity of variance assumption is verified using the Levene's test.

Levene's test for hardness of implants comprising alloy 1 with respect to methods

Null and alternate hypothesis:

Null hypothesis

H_0 : All the population variances are equal

Alternative hypothesis

H_a : At least one variance is different from the rest

Calculate P-value:

Assumption:

Homogeneity of Variance

The P-value is calculated using levene()

The p-value is 0.0034160381460233975

Since the p-value is lesser than the level of significance of 0.05, we reject the null hypothesis of homogeneity of variances.

Insights:

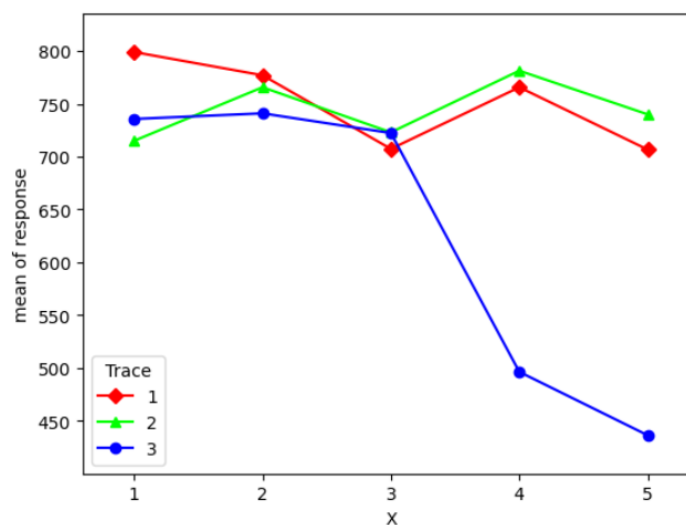
The p-value suggests that the hardness of implant does not follow the homogeneity of variances.

Conclusion:

- Population variances are not equal. Here, we are assuming it to be.
- Sampling is done correctly. Observations for within and between groups must be independent.
- The dependent data must be measured at an interval scale.
- There should be no significant outliers.

4.4.1.3. Calculate P-value:

	df	sum_sq	mean_sq	F	PR(>F)
C(Dentist)	4.0	106683.688889	26670.922222	3.899638	0.011484
C(Method)	2.0	148472.177778	74236.088889	10.854287	0.000284
C(Dentist):C(Method)	8.0	185941.377778	23242.672222	3.398383	0.006793
Residual	30.0	205180.000000	6839.333333	NaN	NaN

**4.4.1.4 Conclusion:**

- The F statistic, that is, the variance between dentist and method interaction is about 3.4 times within each dentist and method interaction.
- The p-value (0.006) is lesser than the level of significance, which implies that the null hypothesis is rejected.
- There are no parallel lines in the plot, implying potential interaction between the dentist and the methods used. This interaction effect indicates that the relationship between dentist and hardness of implant depends on the method of implant used.
- *When dentist 1 employs method 2, the mean response is higher, whereas it drops when method 3 is used.
- When dentist 5 employs method 1, the mean response is higher, whereas it drops when method 3 is used.

- Method 1 shows a higher response with dentist 2, whereas method 3 demonstrates a decrease in response.
- Method 3 shows significantly reduced response when used by dentist 4.
- The lines representing method 1 and method 2 interact with each other however the method 3 does not.
- At dentist 3, all three methods yield similar responses, indicating there might not be substantial interaction among the methods linked to dentist 3.
- We have enough statistical evidence to conclude that there is an interaction effect between dentist and method on the hardness of implants comprising alloy 1.

4.4.2 ALLOY 2

4.4.2.1 Null and alternate hypothesis:

Null hypothesis

H_0 : The hardness of the implants comprising alloy 2 does not depend on dentists and methods for implants.

Alternative hypothesis

H_a : There is an interaction effect between dentist and method on hardness of the implants comprising alloy 2.

4.4.2.2 Checking assumptions of hypothesis test

- The problem concerns with the effect of two independent variables on a dependent variable.
- Two-way ANOVA test is the appropriate test.

Following are the assumptions of the Two-way ANOVA test:

- The populations from which the samples are obtained must be normally distributed.
- The variances among populations must be equal.
- Sampling is done correctly. Observations for within and between groups must be independent.
- The dependent data must be measured at an interval scale.
- There should be no significant outliers.

Shapiro-Wilk's test for hardness of implants comprising alloy 2

Null and alternate hypothesis:

Null hypothesis

H_0 : Hardness of implant follows a normal distribution

Alternative hypothesis

H_a : Hardness of implant does not follow a normal distribution

Calculate P-value:**Assumption:**

Normality

The P-value is calculated using shapiro()

The p-value is 1.1945070582441986e-05

Since p-value of the test is very small than the level of significance of 0.05, we reject the null hypothesis that the hardness of implant follows a normal distribution.

Conclusion:

The p-value suggests that the hardness of implant follows does not follow normal distribution.

Levene's test for hardness of implants comprising alloy 2 with respect to dentists**Null and alternate hypothesis:**

Null hypothesis

H_0 : All the population variances are equal

Alternative hypothesis

H_a : At least one variance is different from the rest

Assumption:

Homogeneity of Variance

The P-value is calculated using levene()

The p-value is 0.23686777576324952

Since the p-value is larger than the level of significance of 0.05, we fail to reject the null hypothesis of homogeneity of variances.

Insights:

The p-value suggests that the hardness of implant follows homogeneity of variances.

Conclusion:

Population variances are equal. The homogeneity of variance assumption is verified using the Levene's test.

Levene's test for hardness of implants comprising alloy 2 with respect to methods

Null and alternate hypothesis:

Null hypothesis

H_0 : All the population variances are equal

Alternative hypothesis

H_a : At least one variance is different from the rest

Assumption:

Homogeneity of Variance

The P-value is calculated using levene()

The p-value is 0.04469269939158668

Since the p-value is lesser than the level of significance of 0.05, we reject the null hypothesis of homogeneity of variances.

Insights:

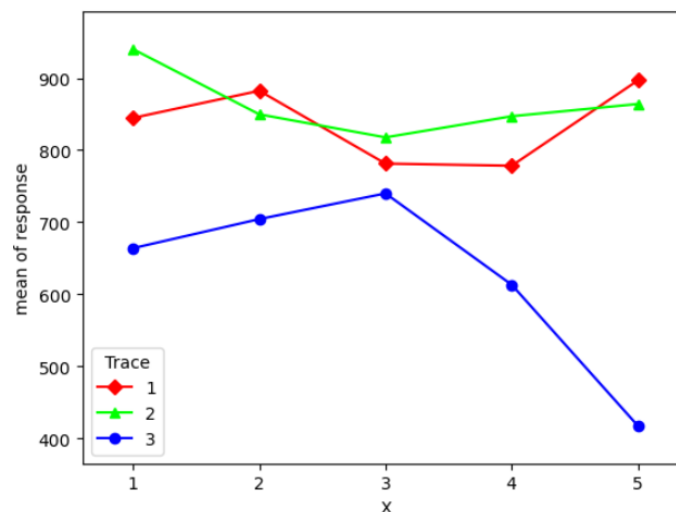
The p-value suggests that the hardness of implant does not follow the homogeneity of variances.

Conclusion:

Population variances are not equal. Here, we are assuming it to be.

4.4.2.3. Calculate P-value:

	df	sum_sq	mean_sq	F	PR(>F)
C(Dentist)	4.0	56797.911111	14199.477778	1.106152	0.371833
C(Method)	2.0	499640.400000	249820.200000	19.461218	0.000004
C(Dentist):C(Method)	8.0	197459.822222	24682.477778	1.922787	0.093234
Residual	30.0	385104.666667	12836.822222	NaN	NaN



4.4.2.4 Conclusion:

- The F statistic, that is, the variance between dentist and method interaction is about 1.9 times within each dentist and method interaction.
- The p-value (0.09) is higher than the level of significance, which implies that we fail to reject null hypothesis.
- There is no evident intersection between all the 3 methods which questions the potential interaction between the dentist and the methods used.
- When dentist 1 employs method 2, the mean response is higher, whereas it drops when method 3 is used.
- When dentist 5 employs method 1, the mean response is higher, whereas it drops when method 3 is used.
- Method 1 shows a higher response with dentist 2, whereas method 3 demonstrates a decrease in response.
- Method 3 shows significantly reduced response when used by dentist 4.
- The lines representing method 1 and method 2 interact with each other however the method 3 does not.
- At dentist 3, all three methods yield similar responses, indicating there might not be substantial interaction among the methods linked to dentist 3.
- We have enough statistical evidence to conclude that there is no interaction effect between dentist and method on the hardness of implants comprising alloy 2.